Analyzing the effect of Railways on price convergence of Rice and Wheat in Colonial India

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Price convergence indicates that prices across different markets become more similar over time, reflecting increased market integration. As prices align, it suggests that markets are responding to the same supply and demand factors, enhancing overall market efficiency and connectivity (Cassidy & Hanley, 2022, pp. 149-174). Market integration can lead to enhanced economic growth, particularly in favorable conditions (Juraev, 2013, p. 18). There is extensive literature documenting the sharp price convergence in the second half of the 19th century in India.

There has been a constant debate about whether the British rule had a positive impact on the Indian economy, there have been various contrasting opinions about whether the technologies introduced by the British contributed to the native Indian economy. Most of these technologies were built to support the infrastructural framework of the British and to exploit Indian resources. The railways were a part of the "new" system and have usually been described as **the** defining feature of India. As Kerr (1995) says: the railways had come to possess India and made its hugeness graspable (p. 1).

Through this study, I hope to analyze price convergence, which in theory is highly likely to be accelerated by the expansion of railways. There have been several papers published that discuss this question using different analysis methods. I hope to quantify and explain the effect of railways on the fall of price dispersion.

The Indian Railways

By the 1850s, railways were being built in many countries across the world, but the East India Company was disinterested in joining this development. "After the almost simultaneous

opening in 1830 of the world's first major railways in England and in the USA, a few progressive Indian businesspeople had suggested creating a network in India" (Wolmar, 2017, p. 9). But these early initiatives like the assist lines for transport of granite at Madras and the lines at a dam project in Rajahmundry proved to be primarily insignificant for the development of the railway network in India.

The increasing pressure from the merchants in British persuaded the British government to consider the introduction of railways. The East India Company vehemently opposed the building of the railways. The Court of Directors sent out a series of reasons for why the railways were not compatible with India to the Governor-General. It included several absurd weather-related arguments, one of them being "the effect of the vertical sun", which they did not explain very precisely. But the key point of disagreement was the terms of the annual dividends that were to be paid to the promoters. The Company's resistance proved to be to no avail and the East India Company begrudgingly accepted the inevitable arrival of railways.

"The very way the railways were constructed makes it clear that its main purpose was to serve the interests of the empire rather than the needs of the Indian economy" (Bandyopadhyay, 2015, p. 167). In 1853, Lord Dalhousie set out in his renowned minute the shape, policies and practices of the Indian railways to the East India Company's Court of Directors. The decision to construct railways was mainly to facilitate army movements. Gradually, there increased a need to integrate the Indian market and open it to British imports. To facilitate this integration, the connection from port cities to internal markets and sources of raw materials was required. "The absence of cheap and reliable transport meant that regions were effectively cut off and unable to trade, greatly limiting the size of potential markets and consequently restricting the potential economic development of the subcontinent" (Wolmar, 2017, p. 33). So, British capital

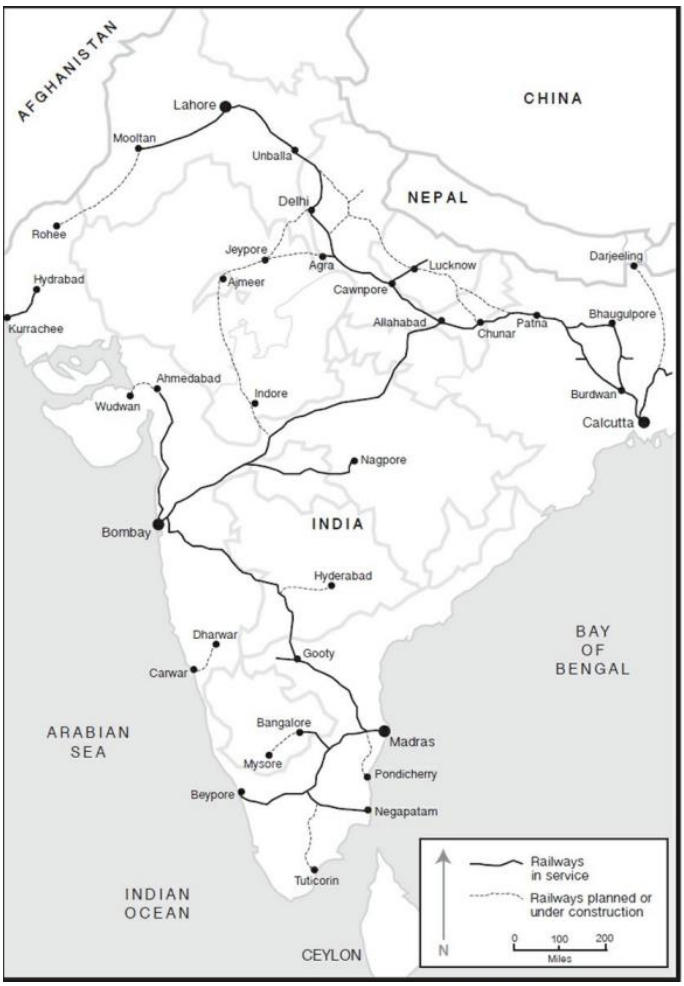
investments (promoters) were invited with 5 percent guaranteed interests to be paid. In 1853 the first commercial train was introduced in Bombay by the Great Indian Peninsula Railways (GIPR). Railways in Calcutta commenced the next year by the East Indian Railways (EIR) and in Madras by the Madras and Southern Mahratta Railway (M&SM). These companies were given free land with a 99-year lease, after which the lines would become government property, but the companies could return the lines to the government during this period of 99 years. "In other words, they could enjoy 5 per cent guaranteed profit for ninety-eight years and then get back all their capital." (Bandyopadhyay, 2015, p. 167). "Overall, eight companies, all created by UK investors, built lines in this first period of Indian railway construction and all benefited from the guarantee system which gave investors a generous rate of return" (Wolmar, 2017, p. 58). This agreement attracted massive investments amounting to £70,110,000.

> The railways grew from the main ports towards the inland areas (indicated in Figure 1) as the main purpose was to tie up the Indian Hinterland to develop foreign trade rather than improve the Indian economy. The inaugural railway line was built by GIPR from Bombay to Thane and EIR was created to connect Calcutta to north India. Major construction projects were underway. Track mileage grew from only 838 miles in 1860 to 4771 miles by 1870 and 24,751 miles by 1900. It grew to about 30,000 miles by 1905 and expanded slowly till independence. (Andrabi & Kuehlwein, 2010, p. 354).

The railways provided several economic benefits to Britain, one of which being the establishment of a dependable supply of cheap food and other goods. The minutes of Dalhousie also discussed this in detail: where one of the main reasons for the establishment would be the exploitation of unexplored sources. They were successful in this venture, the value of Indian exports skyrocketed from £21 million in 1853 to £333 million in 1920.

Figure 1

Railway network in 1871



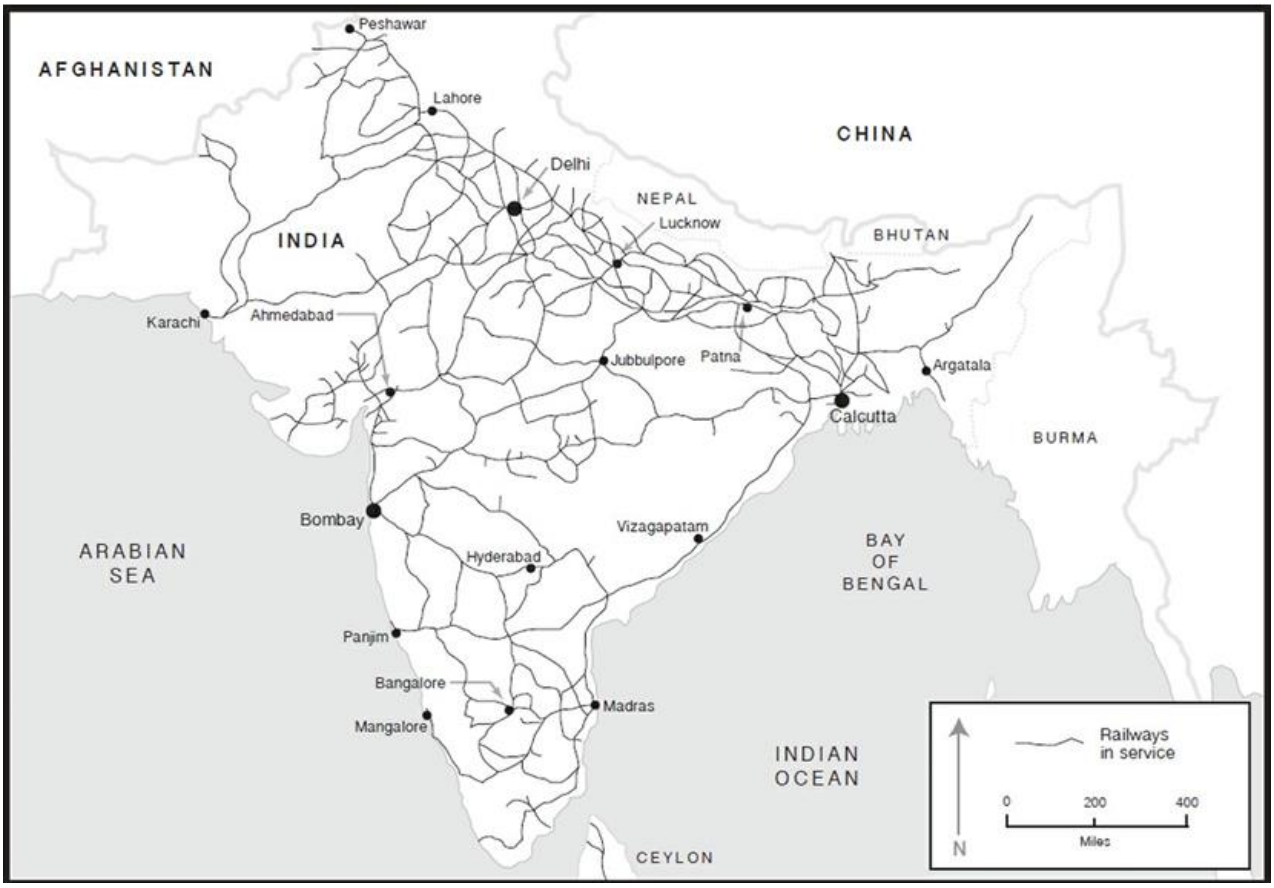
Source: Railways and the Raj: How the Age of Steam transformed India

The government also invested directly in Railway construction. Railway lines were built in the northwest to facilitate troops and supply movements to the frontier. Some official strategic lines were built that stretched to places like Iran and Afghanistan (Andrabi & Kuehlwein, 2010, p. 354). Several lines which were designated as commercial as those to Lahore, Rawalpindi and Multan also had significant military purposes. They also built “famine lines” in scarcity areas. In an 1874 famine, Tirhut was able to receive supplies thanks to a railway connection with Patna.

The concept that railways could be used for famine relief gained interest and the Famine Commission in 1880 suggested that India requires a network of 20,000 miles to be safe from episodes of widespread starvation. The construction of these famine lines was also somewhat limited, as many started to argue that irrigation would have been a more suitable sector to invest in.

Figure 2

Railway network pre-independence



Source: Railways and the Raj: How the Age of Steam transformed India

There were several factors that governed the development of the railway network in India. These play an important role in determining the regions which were able to experience market integration.

The Rice and Wheat Market

"Agricultural production expanded from 1870 to 1900 at an average rate of about 1 per cent per year and cultivated area expanded by 30-40 per cent between 1870 and 1946." (Chaudhary et al., 2016, p. 103). Rice and wheat were the main food grains of India, having approximately 35 and 11 per cent respectively of the net area sown. Bengal specialized in rice production and Punjab in wheat.

Wheat production and consumption during this period were prominent in northern India, mainly the Punjab and United Provinces regions. These regions accounted for 52 per cent of wheat acreage in 1885/86 and grew to 70 per cent in 1919. Most of the wheat was consumed in these areas itself, according to Connell (1885) only 5 per cent of wheat was exported in 1885 (p. 238).

Most rice production and consumption occurred along the east coast in regions like Bengal, Bihar and Madras. Like wheat, rice had 70 per cent acreage in these regions (in 1919) and only 7 per cent export in 1885.

The export of rice and wheat through the railways was massive. In 1909, approximately 30 lakh tons of rice ("in husk" and "not in husk" combined) were exported and in the same year 15 lakh tons of wheat ("flour" and "raw" combined) were exported through the railways. (Administration Report, 1909, pp. 255-256).

The prices of commodities were consistent with the areas they were produced in. The price of rice was generally lower in the rice producing areas of Bengal and Madras and higher in other parts of India. Wheat also followed the same pattern.

The hypothesis/theory is that this price difference would gradually decrease as the railway network in India expands.

Methodology

To analyze the impact of railway expansion on the price convergence of rice and wheat in colonial India, this research will focus on two primary aspects: (1) the correlation between the number of goods conveyed by railways and price convergence, and (2) the disparity in price convergence in districts with railway lines and in districts without lines. The study will focus on the period of 1868-1912.

The data for the number of tons of goods and minerals conveyed for the period of 1867-1912 were obtained from several volumes of *Statistical Abstract relating to British India* (data initially in tons moves on to thousands of tons) and the prices of Rice and Wheat in several districts for the period of 1861-1885 was obtained from the fourth edition of *Prices and Wages in India* (data represented as quantity sold per rupee in seers (1.25 Kg)) and the price data for the following years is obtained from several *Statistical Abstract relating to British India* (data represented as quantity sold per rupee in maund (37.3242 Kg)). The data for occurrence of railways has been estimated by the GIS database which was constructed by Dave Donaldson for his research paper: *Railroads of the Raj: Estimating the impact of Transportation Infrastructure* and the year of arrival of rail lines data present in the *Census of India, 1911*.

The metric used to measure price convergence is Coefficient of Variation (CV_{iy}) where $CV = \frac{\sigma}{\mu}$; σ – Standard Deviation of Prices (Rupees per Seer/ Maund) of *i* crop in *y* year, μ – Mean of the Prices (Rupees per Seer/ Maund) of *i* crop in *y* year. This metric will be useful for comparing data recorded in current prices, meaning the prices at which goods were valued during time of measurement, without adjustments for inflation. As this study analyses economic data over a long period of time, it becomes important to consider inflation and other factors that would affect purchasing power.

For the first section of this study, CV_{iy} will be calculated for both *i* = Rice and Wheat and *y* ranging from 1868-1912 considering data available in *Price and Wages* and *Statistical Abstracts* keeping the regions considered, consistent. This data will be compared with the total number of goods conveyed in the whole nation (GC_y) (in tons). Then the Pearson correlation coefficient(r) is to be calculated. As discussed before, the theory states that r should be negative, implying that as CV decreases--indicating greater price convergence--the growth in trade (GC) should be rising.

The second part of this study, we will calculate CV_{iyr}, the coefficient of variation; *i* could be either rice or wheat and *y* will range from 1868 to 1912, r will either be regions with railways or regions without railways. If railway infrastructure significantly contributed to market integration the CV for districts with railways should be consistently (and significantly) lower than the CV of districts with no railways.

It is also essential to control several factors that could affect price convergence, such as agricultural productivity and several external shocks like lack of rainfall and famines. These

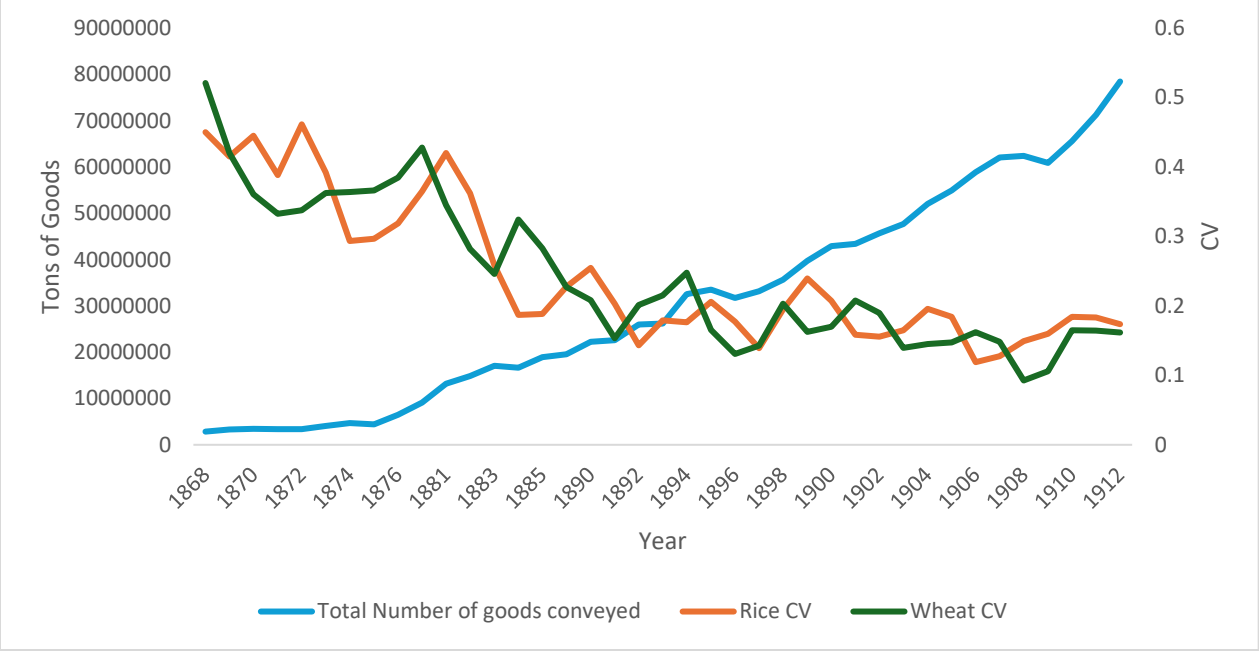
discrepancies, rather than being accounted for in the statistical analysis, have been explained in the *results* section.

Results

Analyzing (1) The correlation between the number of goods conveyed by railways and price convergence.

Figure 3

Total Number of Goods and Coefficient of Variation of Rice and Wheat from 1868-1912



Note. All terms described in text.

The Pearson coefficient for price convergence of rice is -0.7666, indicating a strong negative correlation. This suggests that as the quantity of goods transported increases, the coefficient of variation should decrease, and vice versa, as shown in Figure 1. Similarly, the Pearson coefficient for wheat is -0.826, implying that wheat follows a trend like that of rice. The

lower value of the Pearson coefficient for rice can be attributed to the pre-existing cheap river transport in Bengal. (Wolmar, 2017, p. 33).

One possible way to explain is the difference in cost of transportation: Famine Commission (1880) states that the average price of cart was 2-2.5 annas per ton per mile (1.5 annas in best cases and 4-5 annas in hilly tracts) (as cited in Hurd, 1975, p.266). Meanwhile, the transport costs of railways were 5-6 pies (0.4-0.5 anna) per ton per mile.

This study only accounts for Rice and Wheat crops but according to Hurd (1975) "since they are the largest crops in terms of acreage, it is expected that the pattern they present is indicative of the general pattern for foodgrains." (p. 269). Another important aspect that needs to be considered when interpreting this data is that

> It should not be used to infer that the prices were converging at the same rate in the countryside around the town as all prices were recorded from district headquarter towns. As the cost of moving a seer/ maund by a bullock cart was higher than the cost of moving the same distance by rail, the difference in price of commodities in two neighboring district headquarters connected by rail would be less than between prices between regions in countryside of these towns (McAlpin, 1975, pp. 664–665).

The price convergence for both rice and wheat stagnate after 1905 whereas the number of goods conveyed increases, this can be attributed to the halt in expansion of railways in the middle of the first decade of the 20th century. This part of the study clearly indicates that railways had a defining impact on the sharp price convergence that the agriculture market experienced in the period of 1868-1912, but in this study, we don't consider other factors that

would also affect the fall in price dispersion. To quantify the effect of railways, the second part of the study is required.

Analyzing (2) The disparity in price convergence in districts having railways and the ones not having railways.

The hypothesis that the railways were a driving factor in market integration has been tested. In the case of rice, the coefficient variation in districts with no railways are consistently higher than the CV in districts with railways in the period of 1868-1912 (evident in Figure 3). In only 8 years of this period the CV of districts with railways is higher. The CV of wheat in districts with railways is always higher than the districts without railways.

Figure 4

Coefficient of variation of Rice in railway and non-railway districts.

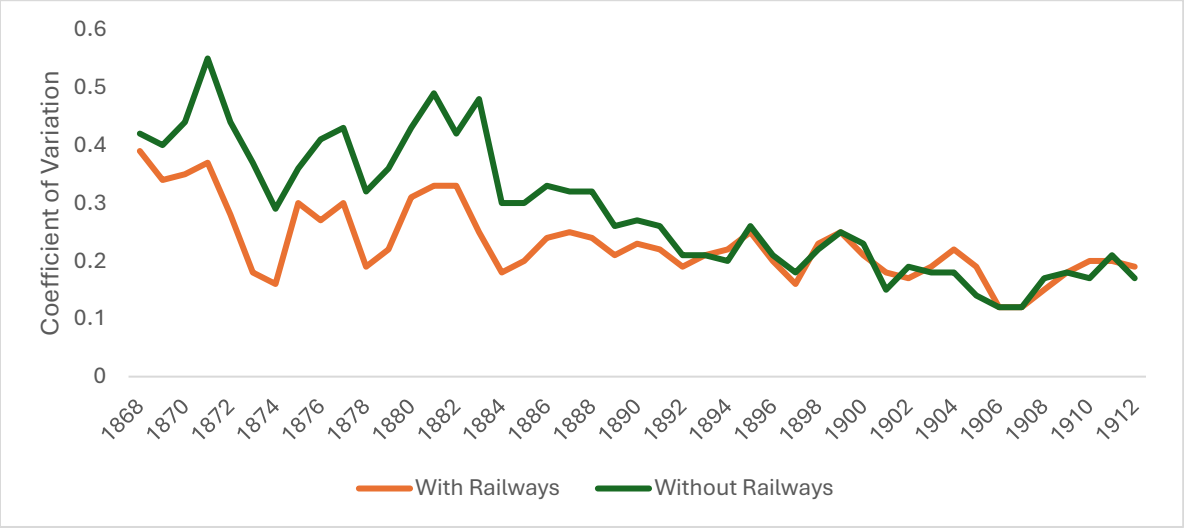
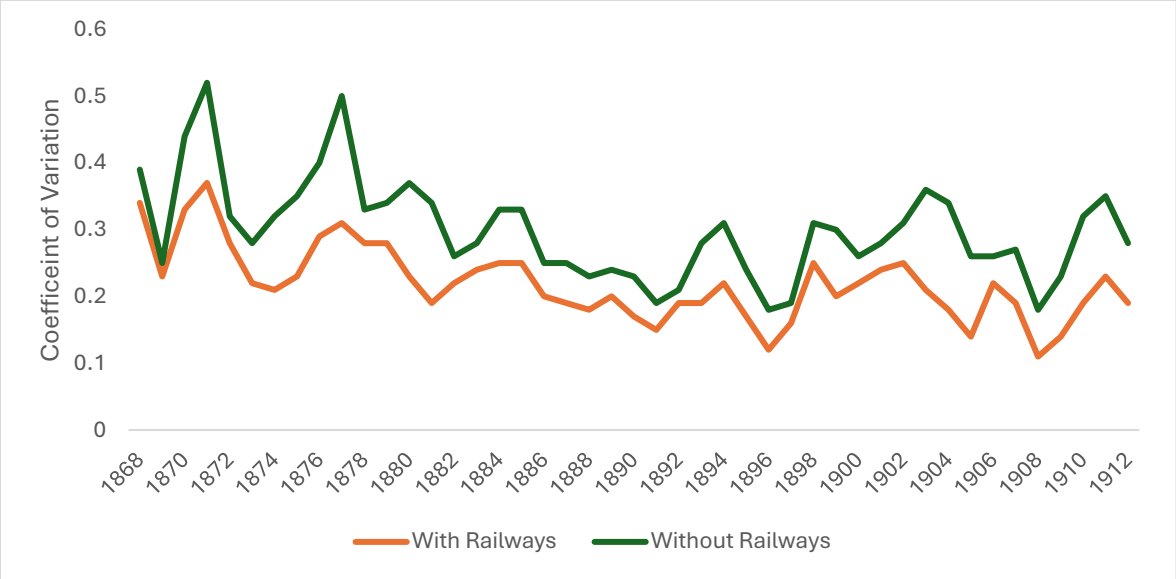


Figure 5

Coefficient of variation of Wheat in railway and non-railway districts.



While the CV was lower in railway districts than in non-railway districts, the pattern of CV through the years remains the same for them. The fluctuations in the trend (peaks and dips) are almost identical in both railway and non-railway districts. Even the magnitude of the differences between the CV of railway and non-railway districts seems marginal.

In non-railway districts, “traders were probably able to utilize lines beyond the district’s borders when railroads and stations were close by.” (Hurd, 1975, p. 275), so as railways expanded in the subcontinent, non-railway districts also integrated into the Indian market. Another reason may be that India was already a partially integrated market due to the pre-existing trade routes like the Grand Trunk Road (opened in 1839) and the river transport through the Ganges River. The railways had also encouraged the construction of feeder roads and roads that interconnected different regions of India. At the same time several infrastructural changes occurred, the telegraph was brought in, and the postal system was standardized. These technological changes also played a part in price convergence of commodities (Andrabi et al., 2019, p. 48). Institutional reforms that were introduced at this time can also be seen as a possible

reason for the fall in price dispersion. Several land reforms were introduced and new education policies like Charles Wood's Education Dispatch were introduced (1854). The currency policy also introduced a single currency (Madras Rupee) all over the British occupied regions. In short, these reforms removed several impediments that existed in the market. After continuous famines in the first three quarters of the 19th century, the Famine Commission was set up that also introduced several provisions for relief work such as "gratuitous relief" or "village dole", in which Doles were distributed to the village people (McAlpin, 1979, p. 151). They encouraged the private sector to provide employment for those people whose employment had been cut off due to lack of rain, drought et cetera. Loans for agriculturists to effect improvements on their farms were made under the Land Improvement Loans Act IX of 1883. These loans were low interest and easily dispensable. All these factors played an important role in market integration.

Conclusion

Through our statistical analysis, it is very clear that there was a significant fall of price divergence in the period of 1868 to 1912. But determining the primary driving factor of this phenomenon is not possible. The (1) part of our analysis just shows that at the same time at which the productivity of railways increased, there was a fall in price divergence. So, it is incorrect to conclude that railways were the primary reason for market integration. The (2) part of our study further concreted the fact that railways were not the primary reason for price convergence.

Other factors that theoretically could have helped in this integration were traders, pre-existing trade routes, introduction of new technologies and several pan Indian institutional frameworks. Railways clearly mattered as the districts with railways had a slightly lower CV

than districts without railways. The current analysis can be easily extended by introducing variables to reduce error and make it more thorough.

This study does not add anything to the question of whether the British institutions brought change to Indian economy or not as it is clear that in the British occupied period, price convergence happened, and it does not matter whether this price convergence happened through British institutions other than the railways. But this study provides an interesting question of how a phenomenon is usually a result of the proceedings of various institutions.

The period of sharp price convergence led to market integration and expansion. Market integration helps align market prices and living standards across regions, reducing economic disparities and promoting more uniform development. While the railways set in motion various political and economic transformations in this country, it is incorrect to assume that they were the driving force behind price convergence.

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